Summer Internship

SIMM851

On

“Full stack Developer”

Completed at:

ThirdEye AI

Submitted by:

Pradeep Swain

2401560056

MCA (2nd year)

K. R. MANGALAM UNIVERSITY, GURUGRAM, HARYANA, INDIA

Training Duration: 1st July 2025 – 31st September 2025

Mode: Full-Time, On-Site

CERTIFICATE

ThirdEye AI

INTERNSHIP CERTIFICATE (Ongoing)

This is to certify that **Mr. Pradeep Swain**, bearing **Roll No. 2401560056**, a student of Master of Computer Applications (MCA) at K.R. Mangalam University, is currently undertaking a six-month Summer Internship at ThirdEye AI, commencing on 1st July 2025 and ongoing, with an expected completion date of 31st September 2025.

During his tenure as a Web Developer Intern in the Product Development Team, Mr. Swain has actively contributed to various client projects, including website redesigns, search engine optimization (SEO) enhancements, and full-cycle web development. His performance reflects:

- Proficient technical skills and expertise
- Effective teamwork and collaboration in a professional environment
- Commitment to adhering to project timelines
- A proactive and solution-oriented approach to challenges

This certificate is issued to support Mr. Swain academic requirements for his ongoing internship report submission.

Signature

Sharad Kumar

Sharad Kumar

Senior Full Stack Developer – ThirdEye AI

JBM House Plot No.9, Institutional Area, Sector 44, Gurgaon, Haryana

DECLARATION

I, Pradeep Swain, a student of MCA at K.R. Mangalam University, hereby declare that the Summer Training Report titled "**Full Stack Development**" is a genuine record of the work that I am currently undertaking during my internship at **ThirdEye AI** from 1st July 2025 to 31st September 2025.

The report has been prepared based on my ongoing learning, assignments, and projects under the guidance of my industry mentor at ThirdEye AI and my faculty supervisor. The information presented in this report is true to the best of my knowledge and has been compiled for academic purposes only.

Since my internship is still in progress, some project details and outcomes are subject to change as I continue to work on the assigned tasks and gain further experience.

Date: _____
Place: _____

Signature of Student

(Pradeep Swain)

ACKNOWLEDGEMENT

I express my sincere gratitude to **K.R. Mangalam University** for integrating industry training into our academic curriculum. This initiative not only bridges the gap between theoretical concepts and real-world applications but also enables students like me to develop skills essential for professional success.

I am deeply thankful to **ThirdEye AI**, a renowned technology solutions provider for the restaurant industry, for offering me the opportunity to work as a **Full stack Developer Intern**. The six-month journey was filled with challenges, learning experiences, and the joy of seeing my work deployed on live client websites.

Special thanks to my mentors and the **Product Development Team** at ThirdEye AI for their constant guidance, code reviews, and encouragement. Their constructive feedback helped me refine my coding practices, adopt industry-standard workflows, and think like a problem solver rather than just a coder.

This journey would not have been possible without the unwavering support of my **family and friends**, who motivated me through late-night coding sessions, tight deadlines, and technical hurdles.

ABSTRACT

This report documents the ongoing summer internship I am pursuing at **ThirdEye AI**, a technology company specializing in providing digital solutions for the food and restaurant industry. The internship, scheduled from **1st July 2025** to **31st September 2025**, is focused on enhancing my practical skills in **Full Stack**.

The primary objective of this internship is to bridge the gap between academic learning and real-world application by engaging in tasks such as designing responsive web pages, optimizing website performance, implementing interactive features, and working with modern web technologies like HTML, CSS, JavaScript, PHP, MySQL, jQuery, and Bootstrap.

The report outlines my learning journey so far, including the introduction to industry workflows, task assignments, and skill development achieved in the initial months. Since the training is still in progress, the report highlights completed modules, ongoing tasks, and planned deliverables that will be incorporated in the final submission.

This experience is helping me gain a deeper understanding of front-end and back-end web development, team collaboration in a professional setting, and industry best practices that will be valuable for my future career.

TABLE OF CONTENTS

1. INTRODUCTION 7

2. COMPANY OVERVIEW 8

3. OBJECTIVES OF THE TRAINING 13

4. TRAINING DETAILS 15

5. TRAINING METHODOLOGY 18

6. PROBLEM-SOLVING APPROACHES 22

7. LEARNING OUTCOMES 24

8. CHALLENGES FACED 26

9. INDUSTRY RELEVANCE OF LEARNED SKILLS 27

10. FUTURE SCOPE IN WEB DEVELOPMENT 28

11. TESTING & DEBUGGING IN WEB DEVELOPMENT 29

CONCLUSION 34

1. INTRODUCTION

The 21st century has witnessed an unprecedented digital revolution, where **web technologies** have transformed the way businesses operate, communicate, and deliver services. In the restaurant industry, this transformation has been particularly significant. From menu displays to online reservations, from digital ordering to customer feedback systems, technology has become an indispensable partner in ensuring seamless customer experiences.

Web Development lies at the heart of this transformation. It involves not just creating visually appealing websites, but also ensuring **functionality, performance, security, and scalability**. Businesses, especially in the food and hospitality sector, are constantly seeking solutions that are fast, responsive, and user-friendly.

Recognizing this trend, I chose to undertake my six-month summer training at **ThirdEye AI**, a market leader in providing **digital solutions to restaurants**. This decision was motivated by:

- The opportunity to work on **real-world, client-facing projects**
- Exposure to **modern frameworks** such as React.js and Tailwind CSS
- Understanding **agile development workflows** in a professional environment
- Learning from experienced industry mentors

This report captures my entire training experience in detail — from onboarding to project delivery, from learning to overcoming challenges. It covers **technical processes**, **project documentation**, **code optimization strategies**, and the **professional growth** I experienced during my tenure.

The purpose of documenting this journey is twofold:

1. To fulfill the academic requirement of my MCA program.
2. To serve as a **personal milestone record**, reflecting my transition from a student to an industry-ready professional.

2. COMPANY OVERVIEW

2.1 About ThirdEye AI

ThirdEye AI is a technology-first company dedicated to transforming the way restaurants operate in the digital era. Founded in 2013 by alumni of the Indian Institute of Technology (IIT) and seasoned industry professionals, ThirdEye AI recognized a gap in the restaurant industry:

While restaurants were exceptional at creating great food and customer experiences in person, many struggled to establish the same level of engagement online.

ThirdEye AI's mission was to fill that gap by building integrated, easy-to-use digital tools that empower restaurants to compete with large-scale food aggregators and maintain direct relationships with their customers.

Over the years, the company has built a strong portfolio of clients, ranging from small cafes to international QSR (Quick Service Restaurant) chains, helping them create and manage their own digital ordering systems, websites, and customer loyalty programs without relying solely on third-party platforms.

2.2 Vision and Mission

Vision:

To be the world's most trusted technology partner for restaurants, enabling them to thrive in a digital-first economy.

Mission:

- To provide affordable, scalable, and innovative solutions tailored to the restaurant industry.
- To empower restaurants with ownership of their brand in the digital space.
- To bridge the gap between offline hospitality excellence and online convenience.

2.3 Core Values

1. Customer-Centricity – Every feature and service is designed with restaurant owners and end-users in mind.
2. Innovation – Continuous experimentation with the latest web technologies, UI/UX principles, and automation tools.
3. Reliability – Ensuring websites and ordering systems have 99.9% uptime.
4. Transparency – Open communication with clients regarding timelines, pricing, and deliverables.
5. Collaboration – Cross-functional teamwork across development, design, sales, and support.

2.4 Organizational Structure

ThirdEye AI operates through specialized teams, each focusing on a particular aspect of the restaurant tech ecosystem.

Organizational Hierarchy:

- CEO / Co-Founders
- Operations Head
- Product Development (PD) Team – *Where I was placed*
    - Frontend Developers
    - Backend Developers
    - UI/UX Designers
    - Quality Assurance (QA) Engineers
- Customer Success Team
- Sales & Marketing
- Support & Maintenance Team

2.5 Key Offerings

ThirdEye AI provides a suite of integrated solutions designed specifically for restaurants:

Product/Service	Description	Tech Stack Commonly Used
Point of Sale (POS)	Software for managing orders, billing, and inventory efficiently.	Java, SQL, APIs
Online Ordering Systems	Direct-to-customer ordering platforms that integrate with payment gateways and delivery partners.	React, Node.js, REST APIs
Custom Website Development	Fully responsive, SEO-friendly websites tailored to the client's brand identity.	HTML, CSS, JavaScript, Tailwind
Loyalty & CRM Tools	Platforms to engage customers through discounts, offers, and rewards programs.	PHP, MySQL, APIs
Restaurant Analytics	Real-time insights into sales, customer behavior, and operational efficiency.	Python, Data Visualization Tools

---

2.6 Client Portfolio

ThirdEye AI caters to both domestic and international brands, including:

- Spice of India – Fine dining Indian restaurant chain
- CakeLounge – Premium bakery brand
- Mad Over Donut (MOD) – Popular dessert franchise
- Burro Blancco – Mexican cuisine outlet
- Krispy Kreme – Global doughnut chain

2.7 Market Position & Competitor Analysis

In the restaurant technology industry, ThirdEye AI competes with both domestic and international players.

Major Competitors:

1. POSist – Strong POS solutions, but limited website customization.
2. Petpooja – Known for POS and restaurant billing, but less focus on customer-facing websites.
3. Zomato / Swiggy Storefronts – Popular among small businesses but charge high commissions and limit brand control.

Competitive Advantage of ThirdEye AI:

- No commission on direct orders — restaurants keep their full revenue.
- High customization — each website is tailored to client branding.
- Full integration — POS, CRM, ordering, and analytics in one ecosystem.
- Dedicated support — 24/7 maintenance for premium clients.

2.8 Technology Stack

ThirdEye AI employs a modern tech stack to ensure speed, scalability, and maintainability:

- Frontend: HTML5, CSS3, Tailwind CSS, JavaScript (ES6+), React.js
- Backend: Node.js, PHP, Express.js
- Database: MySQL, MongoDB
- Version Control: Git, GitHub, GitLab
- Deployment: AWS, Netlify
- Design Tools: Canva, Figma, Adobe XD

4. Learning Outcomes

During my internship at ThirdEye AI, I gained both technical and professional skills:

- Technical Skills:
    - Mastered JavaScript fundamentals and built a strong foundation in web development.
    - Gained practical experience in responsive design using HTML, CSS, and Tailwind CSS.
    - Worked on complex code, understood senior developers' codebases, and navigated intricate CSS structures.
- Problem-Solving Skills:
    - Learned to analyze and solve issues that arose while working on live websites.
    - Improved debugging skills and code optimization techniques.
- Professional Skills:
    - Enhanced my ability to communicate with team members and understand client requirements.
    - Improved time management by handling multiple projects simultaneously.

3. OBJECTIVES OF THE TRAINING

The Summer Training Program at ThirdEye AI was designed to provide MCA students with practical industry exposure in a professional environment, enabling us to apply academic knowledge to real-world scenarios.

The objectives of my six-month training were as follows:

3.1 Academic Objectives

- Application of Theoretical Knowledge:
  To bridge the gap between classroom concepts (HTML, CSS, JavaScript, database management) and their implementation in live client projects.
- Understanding the Software Development Life Cycle (SDLC):
  To observe and participate in various stages — requirement gathering, design, development, testing, deployment, and maintenance.

3.2 Professional Objectives

- Hands-on Experience in Web Development:
  To develop and enhance my skills in modern frontend technologies, including React.js and Tailwind CSS.
- Adoption of Industry Standards:
  To learn coding best practices, project management methodologies (Agile/Scrum), and collaborative workflows using version control systems.
- Client Interaction Skills:
  To understand client requirements, communicate effectively with project managers, and deliver solutions that align with client expectations.

3.3 Personal Development Objectives

- Time Management & Multitasking:
  To manage multiple projects simultaneously, balancing priority tasks and urgent bug fixes.
- Problem-Solving & Analytical Thinking:
  To analyze issues, identify root causes, and provide efficient solutions under time constraints.
- Adaptability & Continuous Learning:
  To quickly adapt to new tools, frameworks, and changing project requirements.

Example:

Objective	Outcome	Example Project
Learn responsive design	Built mobile-first layouts using Tailwind CSS	CakeLounge website
Improve debugging	Fixed layout inconsistencies in React components	Krispy Kreme

---

4. TRAINING DETAILS

The training at ThirdEye AI was well-structured, ensuring a balance between guided learning and independent project execution.
The methodology can be divided into four phases:

4.1 Phase 1 – Onboarding & Orientation (Week 1)

- Company Introduction:
  A detailed presentation on ThirdEye AI's products, client base, and market position.
- Team Introduction:
  Met with members of the Product Development Team, including frontend and backend developers, UI/UX designers, and QA testers.
- Tool Setup:
  Installed necessary software and configured development environments, including:
  - VS Code with React.js extensions
  - Git & GitHub authentication
  - Browser Developer Tools setup
- Access Provisioning:
  Received access to staging servers, design files (Figma), and project documentation.

4.2 Phase 2 – Guided Learning (Weeks 2–4)

- Attended technical sessions on:
  - HTML5 semantic elements
  - CSS3 flexbox, grid, animations
  - JavaScript ES6+ features
  - Tailwind CSS utility classes

    - React component structure and props
- Practiced small internal tasks such as:
    - Updating UI elements
    - Styling forms and navigation menus
    - Integrating basic JavaScript functionality

4.3 Phase 3 – Project Contribution

- Assigned to live client projects under supervision of a senior developer.
- Work allocation through Jira task management tool.
- Daily stand-up meetings (10–15 minutes) to discuss:
    - What was done yesterday
    - Tasks for today
    - Any blockers faced
- Weekly code reviews to ensure:
    - Clean, readable code
    - Consistent use of naming conventions
    - Optimization for performance

4.4 Phase 4 – Independent Execution & Final Evaluation

- Took ownership of smaller projects end-to-end, including:
    - Requirement analysis
    - UI implementation
    - Testing
    - Deployment on staging/live servers

- Final review by mentor, assessing:
    - Technical competence
    - Adherence to timelines
    - Problem-solving ability
- Submission of internship report to HR for certification.

4.5 Work Allocation & Collaboration Tools

During my training, the following tools were used daily:

- Version Control: Git, GitHub (for branching, commits, and pull requests)
- Design Tools: Figma, Adobe XD (for accessing and implementing UI designs)
- Task Management: Jira (tracking tasks, bugs, and deadlines)
- Communication: Slack, Google Meet (daily updates and discussions)
- Testing Tools: Browser DevTools, Lighthouse (for performance audits)

5. TRAINING METHODOLOGY

The training at **ThirdEye AI** was structured to balance theoretical learning with practical application. The approach ensured that trainees not only understood concepts but also developed the ability to apply them in real-world scenarios. The training methodology included:

1. **Induction & Orientation**
    - Introduction to the company's culture, vision, and core values.
    - Understanding the organizational structure and work ethics.
2. **Theory Sessions**
    - Conceptual understanding of core web development principles.
    - Discussions on software development lifecycle, agile methodology, and project workflows.
3. **Hands-on Practical Sessions**
    - Daily coding exercises and mini-projects to strengthen technical skills.
    - Guided implementation of responsive and interactive features.
4. **Code Reviews & Feedback**
    - Regular review sessions by senior developers to improve code quality.
    - Constructive feedback on coding standards, performance optimization, and debugging techniques.
5. **Team Collaboration**
    - Working in teams using Git and GitHub for version control.
    - Communication through Slack and Jira for task management.

---

6. Detailed Learning Outcomes

By the end of this training, I was able to:

- **Frontend Development Skills**
  - Build responsive web pages using **HTML5, CSS3, and JavaScript**.
  - Implement **Bootstrap** and **Tailwind CSS** for faster UI development.
  - Add dynamic interactions with **JavaScript** and **jQuery**.
- **Backend Understanding**
  - Basic knowledge of **PHP** for server-side scripting.
  - Understanding of **MySQL** database management, including creating tables and performing CRUD operations.
- **Version Control**
  - Manage repositories, branches, and commits using **Git**.
  - Collaborate effectively using **GitHub** pull requests and merges.
- **Debugging & Optimization**
  - Use browser developer tools for inspecting and fixing bugs.
  - Apply performance optimization techniques for faster load times.

---

7. Technologies Used

The technologies, tools, and frameworks I worked with during the training included:

Category	Tools / Technologies
Frontend	HTML5, CSS3, JavaScript, jQuery, Bootstrap, Tailwind CSS
Backend	PHP
Database	MySQL
Version Control	Git, GitHub

Category	Tools / Technologies
Collaboration	Jira, Slack
Design Tools	Figma, Adobe Photoshop (basic usage)
Deployment	Netlify, GitHub Pages

---

8. Project Development

During my summer training at **ThirdEye AI**, I contributed to the development of **responsive, user-friendly websites** for clients.

Main Project: Restaurant Management Website

- **Objective:**
  To create a responsive web platform for restaurants to showcase menus, manage online orders, and provide location/contact information.
- **Key Features Developed:**
  1. **Homepage** with dynamic hero banner and call-to-action buttons.
  2. **Menu Page** with categorized dishes, prices, and images.
  3. **Order Form** integrated with backend for online order submission.
  4. **Contact Page** with Google Maps integration and form validation.
- **My Contributions:**
  - Designed **responsive layouts** using Bootstrap Grid System and Tailwind utility classes.
  - Implemented **interactive components** using JavaScript and jQuery.
  - Worked on **form validation** for order and contact forms.
  - Collaborated with backend developers to integrate the order management system.

- **Outcome:**
  The final website was **mobile-friendly, SEO-optimized**, and fully functional for client needs.

9. Tools and Technologies Used

- **Development Tools:** VS Code, Sublime Text, Git & GitHub.
- **Design Tools:** Figma, Adobe XD for UI/UX reference.
- **Version Control:** Git for tracking changes and collaborating.
- **Testing & Debugging:** Chrome DevTools, Postman (for API testing).
- **Deployment:** Familiarity with hosting services and deployment workflows.

10. Evaluation and Feedback

- Weekly performance reviews with mentors.
- Continuous feedback loop to improve code quality and efficiency.
- Final evaluation based on project completion, coding standards, and innovation.

6. PROBLEM-SOLVING APPROACHES

During my internship at ThirdEye AI, I have encountered a variety of technical and functional challenges that required a structured and systematic approach to problem-solving. This methodical process has been critical in addressing issues efficiently and ensuring the delivery of high-quality work. My approach to problem-solving consists of the following steps:

1. Identifying the Issue: The first step in resolving any problem is to fully understand its nature. For instance, when a webpage loaded slowly, I used browser developer tools, such as Chrome DevTools, to analyze network requests, check loading times, and identify performance bottlenecks. Reproducing the issue in a controlled environment allowed me to gather detailed information about the problem's context and symptoms.

2. Root Cause Analysis: After identifying the issue, I conducted a thorough analysis to pinpoint its root cause. This involved examining the codebase, reviewing console logs for errors, and tracing dependencies to understand how different components interacted. For example, if an API call failed, I would verify the request parameters and response data to determine whether the issue stemmed from the front-end or back-end.

3. Brainstorming Solutions: Once the cause was identified, I explored multiple potential solutions. I collaborated with my mentor and senior developers to discuss the feasibility, scalability, and maintainability of each option. This collaborative approach ensured that the chosen solution aligned with the project's goals and adhered to best practices.

4. Implementing Fixes: After selecting the most appropriate solution, I updated the code while adhering to the team's coding standards. I ensured that my changes were well-documented with clear comments to facilitate future maintenance. For instance, when optimizing a webpage's performance, I implemented techniques like lazy loading and minified CSS/JavaScript files.

5. Testing After Fix: Post-implementation, I conducted comprehensive testing to verify that the issue was resolved without introducing new bugs. This included functional testing to ensure the feature worked as intended and responsive testing to confirm compatibility across different devices and browsers.

6. Documenting the Solution: To support future debugging and knowledge sharing, I documented each solution in the team's internal documentation system. This included a description of the issue, the root cause, the implemented fix, and any lessons learned. This practice helped streamline troubleshooting for similar issues in the future.

This structured approach enabled me to address a wide range of challenges, including user interface (UI) bugs, performance bottlenecks, and API integration errors, while maintaining high standards of quality and efficiency.

7. LEARNING OUTCOMES

My ongoing internship at ThirdEye AI has been a transformative experience, equipping me with both technical and professional skills that are essential for a career in web development. The key learning outcomes are categorized as follows:

Technical Skills:

- Enhanced proficiency in front-end technologies such as HTML, CSS, JavaScript, jQuery, and Bootstrap. These skills enabled me to build responsive and visually appealing websites that meet modern design standards.

- Gained hands-on experience in integrating APIs to fetch and display dynamic data, such as product listings or user profiles, and updating UI components in real-time.

- Improved debugging capabilities using tools like Chrome DevTools, which allowed me to analyze network activity, inspect DOM elements, and resolve JavaScript errors efficiently.

- Learned web performance optimization techniques, including lazy loading of images, compressing assets, and minifying code, to improve page load times and enhance user experience.

Professional Skills:

- Developed strong time management skills by balancing multiple tasks, such as feature development and bug fixes, while adhering to project deadlines.

- Strengthened collaboration skills through participation in daily stand-up meetings, where I provided updates on my progress and discussed challenges with the team.

- Gained experience in peer code reviews, which improved my ability to write clean, maintainable code and incorporate constructive feedback.

- Enhanced adaptability by quickly learning new tools, workflows, and technologies as required by different projects.

These outcomes have not only deepened my technical expertise but also improved my ability to work effectively in a professional environment, communicate ideas clearly, and approach challenges with confidence.

8. CHALLENGES FACED

Throughout my internship, I encountered several challenges that tested my technical and problem-solving abilities. Overcoming these obstacles has been instrumental in my growth as a developer. The key challenges include:

- Complex Codebases: Initially, navigating large and complex codebases was daunting. Understanding the existing code structure, identifying relevant modules, and adhering to the team's coding conventions required significant effort and time.

- Cross-Browser Compatibility Issues: Ensuring consistent performance across browsers like Chrome, Firefox, and Safari was challenging. For example, some CSS styles rendered perfectly in Chrome but required adjustments to work correctly in Safari.

- API Integration Errors: At times, API responses did not match the expected format, leading to front-end rendering issues. Troubleshooting these errors involved close collaboration with the back-end team to verify data structures and endpoints.

- Time Constraints: Balancing urgent feature updates with ongoing development tasks often required working under tight deadlines, which tested my time management and prioritization skills.

- Debugging Deeply Nested Functions: Tracing bugs in large projects with interdependent functions was time-consuming and required careful analysis to avoid introducing new issues.

By addressing these challenges, I developed resilience, improved my technical depth, and learned to approach problems with a calm and methodical mindset.

9. INDUSTRY RELEVANCE OF LEARNED SKILLS

The skills I have acquired at ThirdEye AI are directly applicable to the web development industry, where demand for efficient, user-friendly, and scalable applications continues to grow. The relevance of these skills includes:

- Responsive Web Design: With the increasing use of mobile devices, creating websites that adapt seamlessly to different screen sizes is a critical industry requirement.

- API Integration Skills: Modern web applications rely heavily on APIs to deliver dynamic content, making proficiency in API integration essential for building interactive platforms.

- Debugging and Testing Practices: Thorough debugging and testing ensure the delivery of stable, high-quality applications, which is a priority for both startups and enterprises.

- Version Control (Git): Using Git for version control is a standard practice in the industry, enabling efficient code management and team collaboration.

- Performance Optimization: Optimizing page load times and resource usage directly impacts user retention, SEO rankings, and business outcomes, making it a highly valued skill.

These skills align with industry demands and position me to contribute effectively to web development projects in diverse professional settings.

10. FUTURE SCOPE IN WEB DEVELOPMENT

Web development is a dynamic field with continuous advancements, and the skills I have gained at ThirdEye AI provide a strong foundation for future growth. My plans for further development include:

- Exploring Modern Front-End Frameworks: I aim to learn frameworks like React.js, Next.js, and Vue.js to build scalable, component-based user interfaces that enhance user experience.

- Full-Stack Development: Expanding my knowledge to include back-end technologies like Node.js, Express.js, and MongoDB will enable me to become a MERN stack developer, capable of handling both front-end and back-end tasks.

- Progressive Web Apps (PWAs): I plan to explore PWAs to create applications that offer offline functionality and a native app-like experience, catering to modern user expectations.

- AI and Automation Integration: Leveraging AI APIs and automation tools will allow me to enhance website functionality, such as adding chatbots or personalized content recommendations.

- Continuous Learning and Upskilling: Staying updated with emerging trends, such as new JavaScript frameworks, web security practices, and UI/UX design principles, will ensure my skills remain relevant.

By pursuing these areas, I aim to build a versatile and future-ready career in web development, contributing to innovative and impactful projects.

11. TESTING & DEBUGGING IN WEB DEVELOPMENT

Testing and debugging are critical stages in the web development lifecycle, aimed at ensuring that a website or web application performs as expected, adheres to client requirements, and delivers a seamless user experience. At ThirdEye AI, I had the opportunity to work with both manual and automated testing techniques to identify, isolate, and resolve potential issues before deployment.

6.1 Importance of Testing

Testing is not just about finding bugs—it's about verifying that the application works as intended under various conditions. Proper testing:

- Ensures **functional accuracy** of features.
- Identifies **performance bottlenecks** before users experience them.
- Helps maintain **cross-browser compatibility**.
- Improves the **security** and **stability** of the application.

6.2 Types of Testing Used

During my training, I worked on several types of testing:

a) Functional Testing

- Verified that all website features (forms, navigation menus, API calls, etc.) worked as per the requirements.
- Example: Testing if the "Add to Cart" button on a food ordering page correctly updated the cart quantity.

b) Responsive & Cross-Browser Testing

- Ensured layouts were consistent across **Chrome, Firefox, Safari, and Edge**.

- Used **Chrome DevTools** to simulate various screen sizes, including mobile devices and tablets.

c) Performance Testing

- Checked website loading speed using **Google Lighthouse**.
- Optimized **images, scripts, and CSS** to improve load time.

d) Accessibility Testing

- Verified compliance with **WCAG standards** for color contrast, keyboard navigation, and screen reader support.

e) Security Testing

- Checked for vulnerabilities like **SQL Injection** or **Cross-Site Scripting (XSS)**.
- Sanitized user inputs before storing or processing.

6.3 Debugging Practices

Debugging was a daily part of my role. I frequently used:

- **Chrome DevTools**: To inspect DOM elements, monitor console logs, and debug JavaScript errors.
- **Network Tab**: To analyze API request/response data and troubleshoot failed requests.
- **Breakpoints in JavaScript**: To stop code execution and check variable values step-by-step.
- **Error Tracking Tools**: Such as **Sentry** for real-time error monitoring.

Example:
While integrating a payment gateway, the checkout process failed due to a mismatched API endpoint. By checking the **Network Tab**, I discovered the staging API URL was still in use instead of the production URL. Updating it resolved the issue.

6.4 Tools Used

- **Postman** – For API testing.
- **Google Lighthouse** – For performance and accessibility audits.
- **BrowserStack** – For cross-browser and cross-device testing.
- **Git** – To manage versions and roll back changes if a bug appeared after deployment.

6.5 Best Practices Learned

- Test early and often instead of waiting until the end of development.
- Write **descriptive console logs** for easier debugging.
- Keep code modular to isolate issues quickly.
- Reproduce bugs before fixing to ensure they are truly resolved.
- Maintain a **bug tracking sheet** to document issues and solutions.

4. Training Objectives

The primary objectives of my summer training at **ThirdEye AI** were:

1. **Practical Application of Academic Knowledge**
   To bridge the gap between theoretical concepts learned during my BCA/MCA coursework and their real-world applications in the IT industry.
2. **Skill Enhancement**
   To gain hands-on experience with web technologies such as HTML5, CSS3, JavaScript, PHP, MySQL, and modern frameworks, enabling me to create responsive, user-friendly, and scalable websites.
3. **Understanding Industry Practices**
   To learn about professional workflows, version control systems (Git/GitHub), agile methodologies, and collaborative project management tools.
4. **Project-Based Learning**
   To contribute to live projects, enhancing my problem-solving skills and ability to work in a team environment.

---

5. Training Methodology

The training program at ThirdEye AI was structured to provide both **theoretical guidance** and **practical exposure**. The methodology adopted included:

1. **Orientation Session**
    - Introduction to the company, its products, and development workflow.
    - Overview of ongoing projects and technologies used.
2. **Technical Workshops**
    - Hands-on sessions on HTML5, CSS3, JavaScript ES6+, Bootstrap, and responsive design principles.
    - Introduction to PHP, MySQL, and database integration.
    - Sessions on Git & GitHub for version control.
3. **Mentor Guidance**
    - Regular one-on-one interactions with senior developers for code reviews and best practices.
    - Feedback on project implementation.
4. **Live Project Work**
    - Assigned to an active project team, where I contributed to front-end and back-end development.
    - Exposure to debugging, optimization, and cross-browser compatibility testing.
5. **Self-Learning & Research**
    - Encouraged to explore documentation, open-source tools, and coding best practices.
    - Applied learnings to solve real-world issues in the assigned project.

---

6. Work Experience / Tasks Performed

During my training, I had the opportunity to work on multiple modules within different projects. My primary responsibilities included:

6.1 Front-End Development

- Developed responsive and interactive web pages using **HTML5**, **CSS3**, and **JavaScript**.
- Implemented UI components such as custom navigation bars, sliders, and forms.
- Used **Bootstrap** and **Tailwind CSS** for styling and responsive layouts.
- Ensured **cross-browser compatibility** and mobile responsiveness.

6.2 Back-End Development

- Worked with **PHP** for server-side scripting.
- Designed and integrated **MySQL** databases for storing dynamic content.
- Implemented **CRUD** operations for managing application data.

6.3 API Integration

- Integrated third-party APIs for payment gateways and location-based services.
- Used **AJAX** and **Fetch API** for asynchronous data loading.

6.4 Version Control & Collaboration

- Used **Git** for tracking changes and maintaining code history.
- Collaborated with team members via **GitHub** pull requests and code reviews.

6.5 Testing & Optimization

- Performed debugging using browser developer tools.
- Improved website performance by optimizing images, reducing load time, and minifying code.

CONCLUSION

The summer training at **ThirdEye AI** has been an enriching and practical learning experience, allowing me to bridge the gap between theoretical knowledge and real-world application. Through active participation in various stages of the web development lifecycle — from designing and coding to testing and debugging — I gained hands-on exposure to industry-standard tools, frameworks, and workflows.

Working alongside experienced professionals helped me understand the importance of clean, maintainable code, effective collaboration, and meeting client requirements within given deadlines. The testing and debugging phase, in particular, reinforced my problem-solving skills and taught me the value of precision, patience, and attention to detail.

Since my internship is still ongoing, I look forward to gaining deeper insights into advanced development practices, enhancing my technical skills, and contributing further to meaningful projects. This experience has strengthened my passion for web development and will serve as a strong foundation for my future career in the IT industry.